Game Engine Concepts

Fachhochschule Oberösterreich

Agenda

1. Overview & Motivation
2. Game Engines
3. Game Loop & Render Pipeline
4. Graphics, Animation, Physics
5. Human Interface Devices
6. Game Scripting

about me

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Fachhochschule Oberösterreich

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Introduction: Overview & Motivation

Overview

The Game Developer’s Mind

- A multi-level architecture
- Users only see the tip of the iceberg

Overview

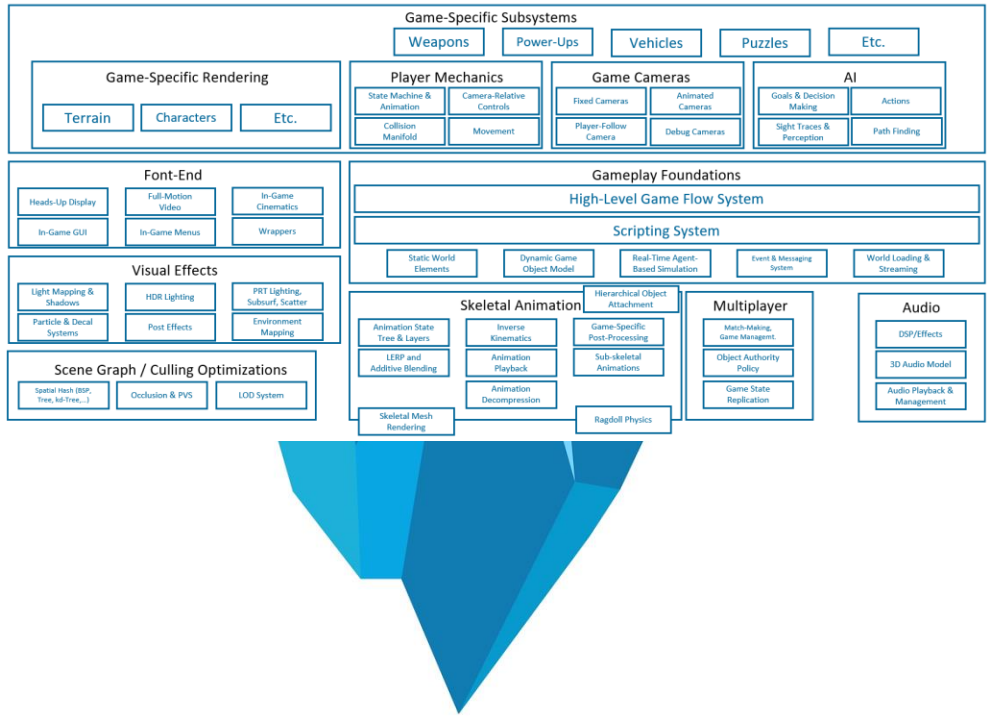
The Game Developer’s Mind

- Fully rendered 3D game scene with interactive components
- Beneath lies a 3D game engine

Overview

The Game Developer's Mind

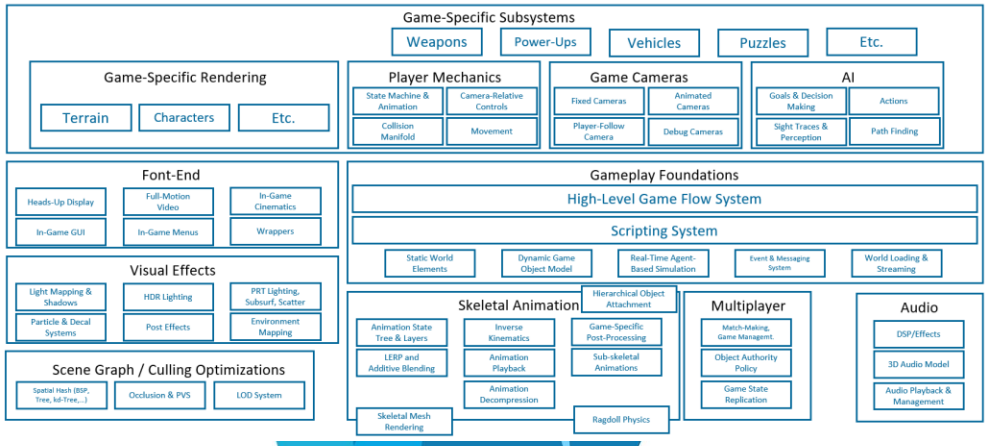
- The game engine is a software framework that separates between core components
- Graphics, game mechanics, physics/collisions, front-end, visual effects, scene graphs, etc.
- Components communicate and act based on code



Overview

The Game Developer's Mind

- The code is nothing else than subsequent instructions in a given programming language
- For a computer scientist, the language doesn't matter, because the concepts stay the same

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// Called to bind functionality to input
void APlayerPawn::SetupPlayerInputComponent(UInputComponent* PlayerInputComponent)
{
    Super::SetupPlayerInputComponent(PlayerInputComponent);

}
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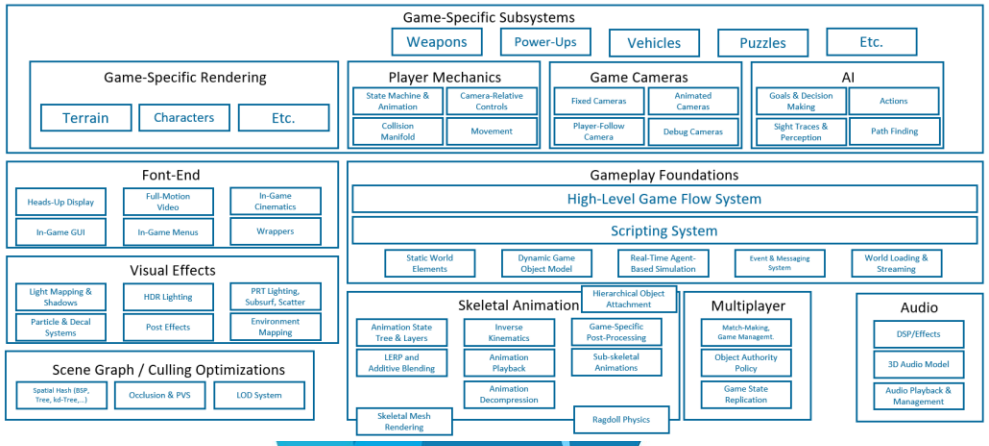
Overview

The Game Developer's Mind

- The programs themselves are translated into machine code
- The machine code is the atomic language of a particular computer
- Computers are both complex and entirely simple

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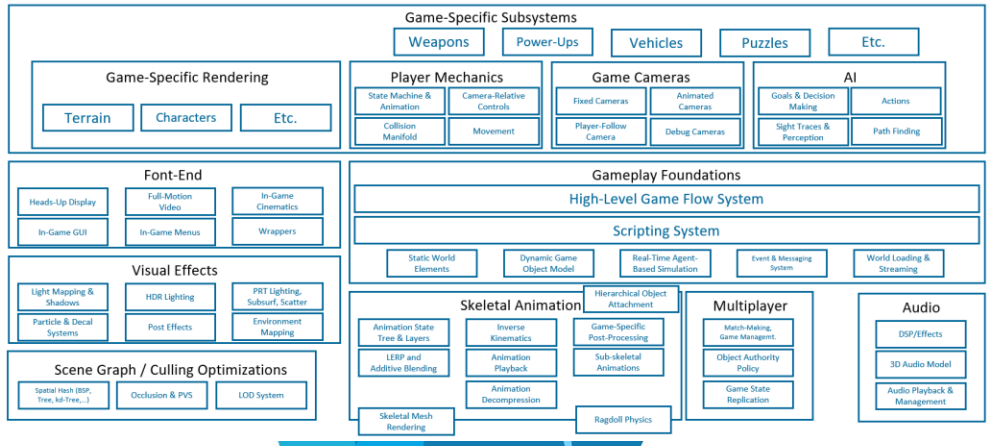
Overview

The Game Developer's Mind

- The programs themselves are translated into machine code
- The machine code is the atomic language of a particular computer
- Computers are both complex and entirely simple
- **Over the semester, we will learn about these concepts on hand with the Unity and Godot Game Engine, starting from the most atomic principles.**

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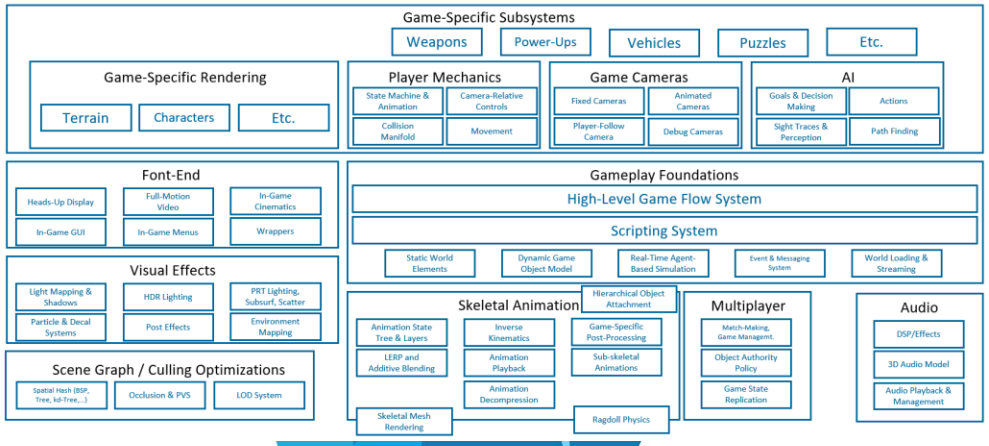
Overview

The Game Developer's Mind

- The programs themselves are translated into machine code
- The machine code is the atomic language of a particular computer
- Computers are both complex and entirely simple
- **However, the focus of this first-semester lecture is not the programming aspect, but rather the concepts needed to use game engines for creating games and simulations, etc.**

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Overview

The Game Developer's Mind

- The programs themselves are translated into machine code
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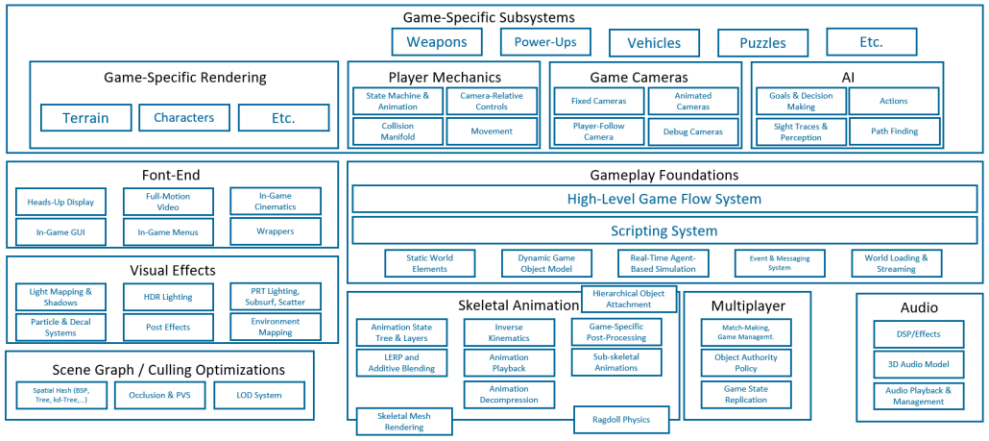
- Computers are both complex and entirely simple
- **Today, we will learn the foundations about game engines**

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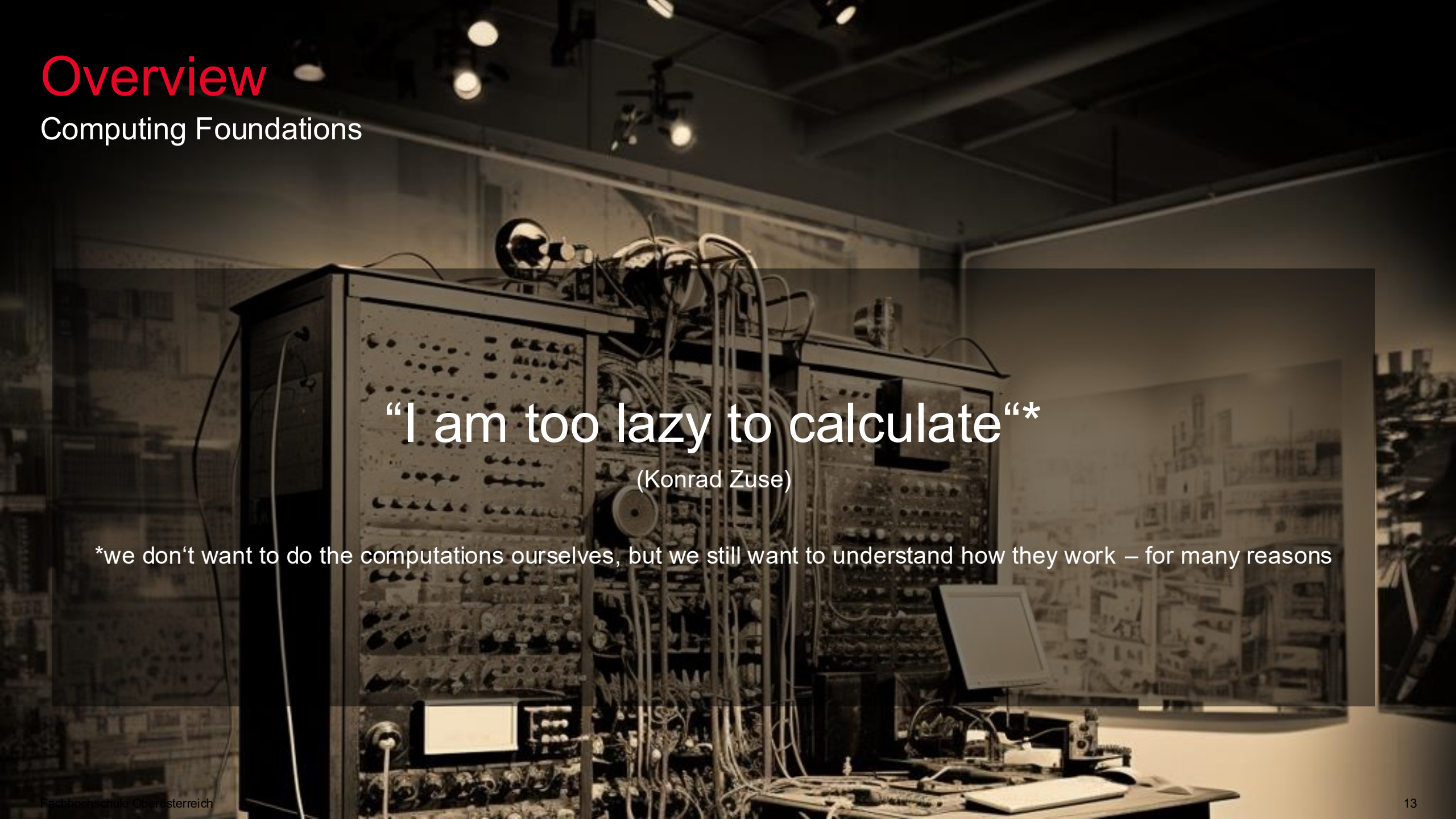
Overview

Computing Foundations



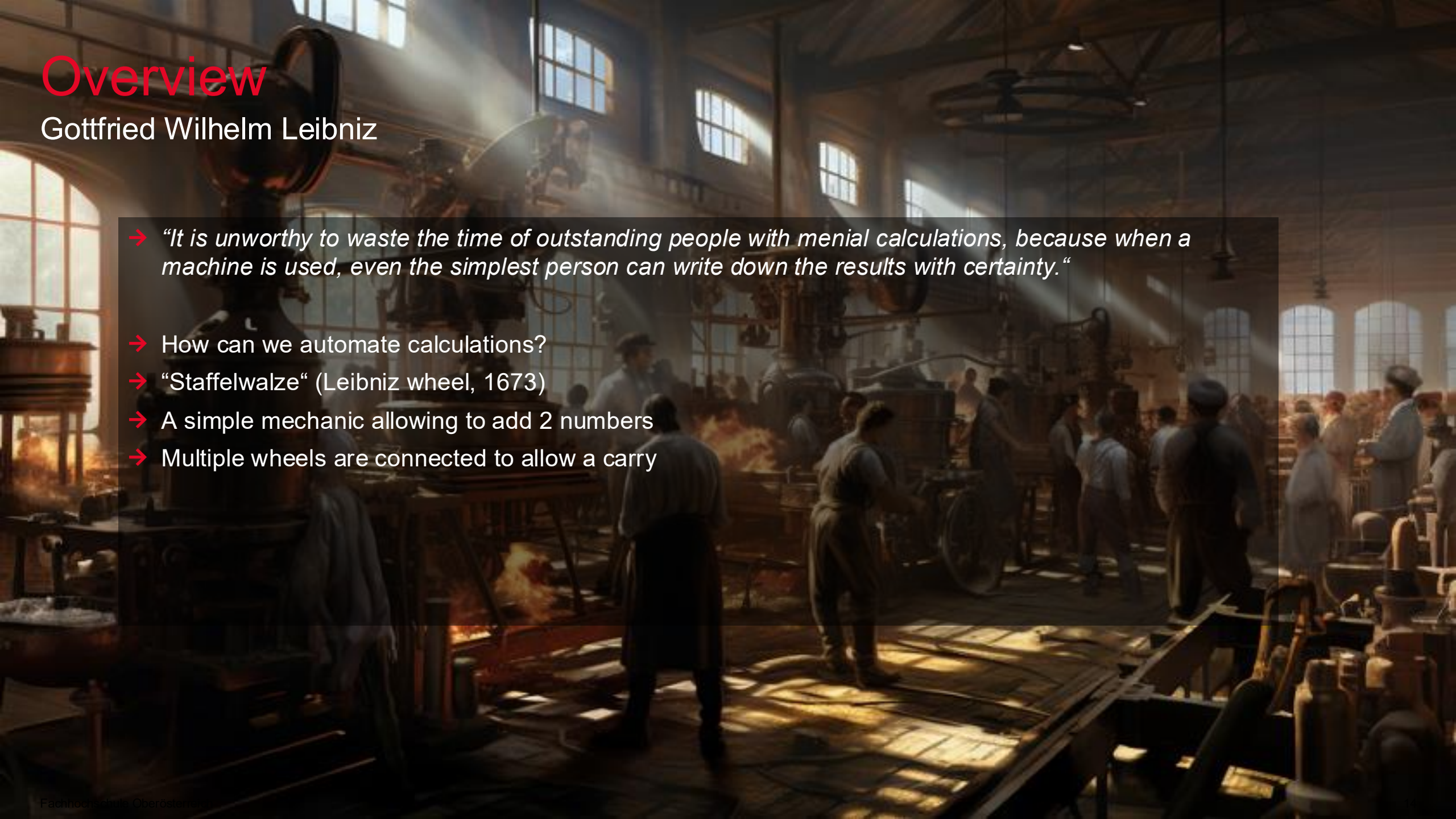
“I am too lazy to calculate“*

(Konrad Zuse)

*we don‘t want to do the computations ourselves, but we still want to understand how they work – for many reasons

Overview

Gottfried Wilhelm Leibniz



- *“It is unworthy to waste the time of outstanding people with menial calculations, because when a machine is used, even the simplest person can write down the results with certainty.“*

- How can we automate calculations?
- “Staffelwalze“ (Leibniz wheel, 1673)
- A simple mechanic allowing to add 2 numbers
- Multiple wheels are connected to allow a carry


ORIGINAL-ODHNER

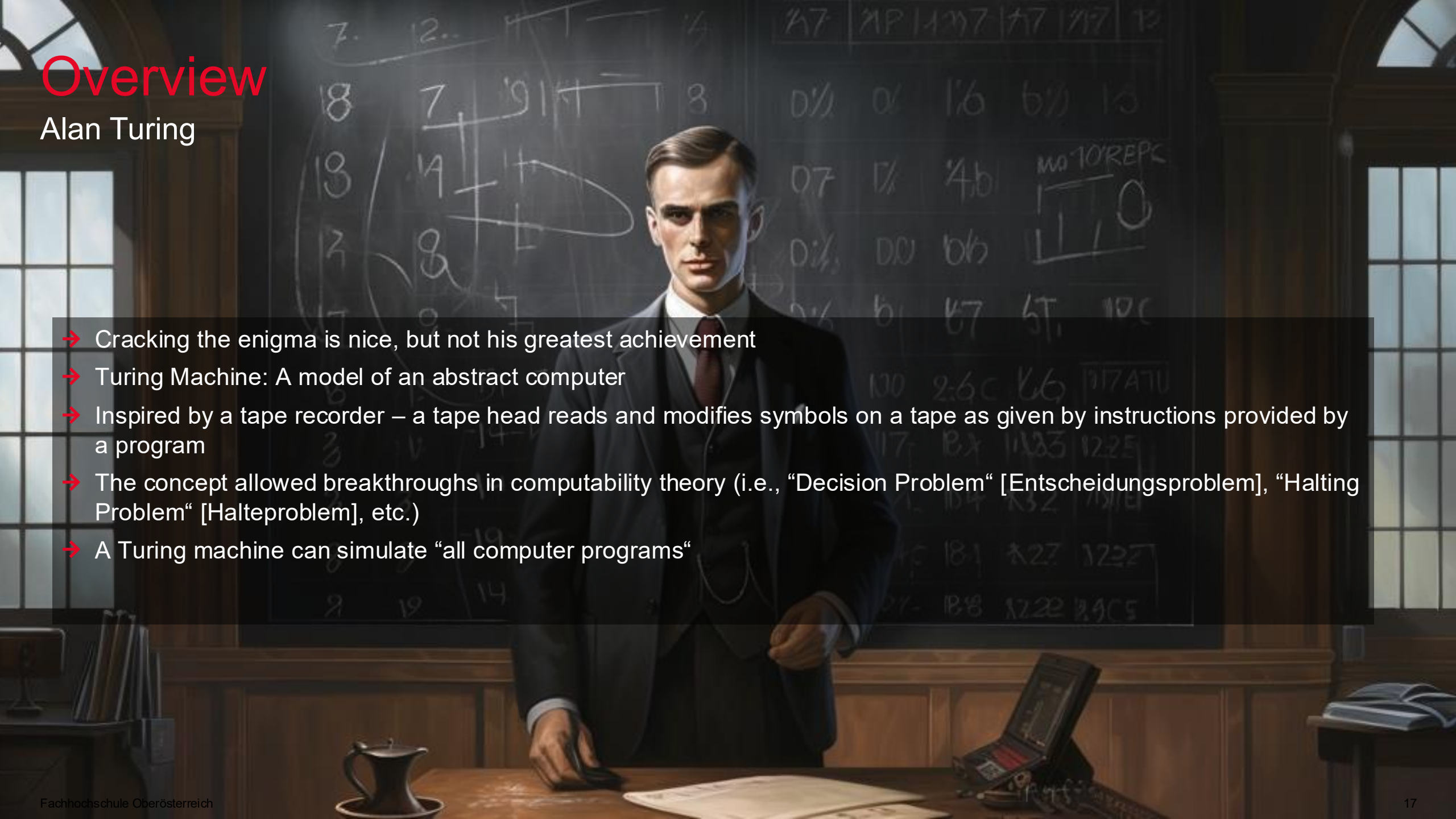
Overview

Konrad Zuse

- First binary computer (Z1, 1938)
- Boolean logic and floating point numbers
- 8 instructions, between 1 and 21 cycles per instruction
- 16 word floating point memory
- Destroyed in WWII, but rebuilt by Zuse himself at the German Technikmuseum in Berlin
- First fully digital computer (Z3, 1941)

Overview

Alan Turing



- Cracking the enigma is nice, but not his greatest achievement
- Turing Machine: A model of an abstract computer
- Inspired by a tape recorder – a tape head reads and modifies symbols on a tape as given by instructions provided by a program
- The concept allowed breakthroughs in computability theory (i.e., “Decision Problem“ [Entscheidungsproblem], “Halting Problem“ [Halteproblem], etc.)
- A Turing machine can simulate “all computer programs“

Overview

Programming

- A program is just a series of subsequent instructions
- To be “Turing complete“ (and thus, allow any program), a language needs to…
  - Have some memory to read from and write to
  - The possibility to perform subsequent instructions
  - A conditional jump (“if“)

Overview

The Game Developer’s Mind

- A game is “just“ a computer program
- To create a game, we should understand the **underlying concepts and mechanics**

Course Modalities

Overview

Course Modalities

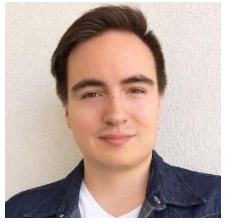
- **Game Engine Concepts (GEC)**
  - 12 EH VO, 12 EH UE
  - Unity Engine, Godot Game Engine
  - Exam: December 16, 2025
  - Tutor: Tobias Grantl (MTD 5. Semester)

Game Engine Concepts

FH Oberösterreich
University of Applied Sciences Upper Austria

Digital Media Department

Media Technology & Design
Digital Arts
Interactive Media



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